



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta preliminare del 28/06/2018*

PROBLEM - We observe N samples of a random process: $X[n] = s[n] + W[n]$, $n = 0, 1, 2, \dots, N-1$, where $s[n] = (-1)^n A$ is a deterministic signal with alternating sign, A is a deterministic unknown parameter that we want to estimate, $W[n]$ is a white Gaussian process with power σ_w^2 .

1) Derive the joint probability density function (pdf) of the N observed data samples $\mathbf{X} = [X[0] \ X[1] \ \dots \ X[N-1]]^T$.

Under the assumption that the noise power σ_w^2 is a-priori known:

- (2) Derive the Maximum Likelihood (ML) estimator of A given the observation of $\mathbf{X}$.
- (3) Derive the bias, the mean square error (MSE) and the pdf of $\hat{A}_{ML}$ and establish if $\hat{A}_{ML}$ is consistent and efficient.

Under the assumption that the noise power σ_w^2 is also unknown:

- (4) Derive the joint ML estimator of A and σ_w^2 and calculate the bias of $\hat{\sigma}_{wML}^2$.



Solution

(1) Joint pdf of the observed data:

$$X[n] \in N((-1)^n A, \sigma_w^2), \quad \{X[n]\}_{n=0}^{N-1} \text{ independent} \Rightarrow \mathbf{X} \in N(\mathbf{A}\mathbf{s}, \sigma_w^2 \mathbf{I}), \quad \text{where } \mathbf{s}_n = (-1)^n.$$

pdf in scalar format:

$$f_x(\mathbf{x}) = \prod_{n=0}^{N-1} f_x(x[n]) = \left(\frac{1}{\sqrt{2\pi\sigma_w^2}} \right)^N \exp \left(-\frac{1}{2\sigma_w^2} \sum_{n=0}^{N-1} (x[n] - (-1)^n A)^2 \right)$$

pdf in vector format:

$$f_x(\mathbf{x}) = \frac{1}{(2\pi)^{N/2} \sqrt{|\sigma_w^2 \mathbf{I}|}} \exp \left(-\frac{1}{2} (\mathbf{x} - \mathbf{A}\mathbf{s})^T (\sigma_w^2 \mathbf{I})^{-1} (\mathbf{x} - \mathbf{A}\mathbf{s}) \right) = \frac{1}{(2\pi)^{N/2} (\sigma_w^2)^{N/2}} \exp \left(-\frac{1}{2\sigma_w^2} \|\mathbf{x} - \mathbf{A}\mathbf{s}\|^2 \right)$$

Log-likelihood function:

$$\begin{aligned} \ln L(A) = \ln f_x(\mathbf{x}) &= -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln(\sigma_w^2) - \frac{1}{2\sigma_w^2} \sum_{n=0}^{N-1} (x[n] - (-1)^n A)^2 \\ &= -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln(\sigma_w^2) - \frac{1}{2\sigma_w^2} \|\mathbf{x} - \mathbf{A}\mathbf{s}\|^2 \end{aligned}$$

(2) Under the assumption that the noise power is a-priori known $\rightarrow$ ML estimate of A:

$$\frac{d \ln f_x(\mathbf{x})}{dA} = \frac{1}{\sigma_w^2} \sum_{n=0}^{N-1} (-1)^n (x[n] - (-1)^n A) = 0 \Rightarrow \hat{A}_{ML} = \frac{\sum_{n=0}^{N-1} (-1)^n x[n]}{\sum_{n=0}^{N-1} (-1)^{2n}} = \frac{1}{N} \sum_{n=0}^{N-1} (-1)^n x[n] = \frac{1}{N} \mathbf{s}^T \mathbf{x}$$

(3) Since the estimate is a linear function of Gaussian data, the estimate is also a Gaussian random variable, with mean and variance given by:

$$E\{\hat{A}_{ML}\} = \frac{1}{N} \sum_{n=0}^{N-1} (-1)^n E\{x[n]\} = \frac{1}{N} \sum_{n=0}^{N-1} (-1)^n (-1)^n A = A \cdot \frac{1}{N} \sum_{n=0}^{N-1} (-1)^{2n} = A$$

Hence, the estimator is unbiased.

$$MSE\{\hat{A}_{ML}\} = \text{var}\{\hat{A}_{ML}\} = \text{var}\left\{ \frac{1}{N} \sum_{n=0}^{N-1} (-1)^n x[n] \right\} = \frac{1}{N^2} \sum_{n=0}^{N-1} (-1)^{2n} \text{var}\{x[n]\} = \frac{\sigma_w^2}{N}$$

Hence, the estimator is consistent.



$$\frac{d^2 \ln f_X(\mathbf{x})}{dA^2} = \frac{d}{dA} \left(\frac{1}{\sigma_w^2} \sum_{n=0}^{N-1} (-1)^n (x[n] - (-1)^n A) \right) = -\frac{1}{\sigma_w^2} \sum_{n=0}^{N-1} (-1)^{2n} = -\frac{N}{\sigma_w^2}$$

$$CRB(A) = \frac{1}{-E \left\{ \frac{d^2 \ln f_X(\mathbf{x})}{dA^2} \right\}} = \frac{\sigma_w^2}{N} = MSE \{ \hat{A}_{ML} \}$$

Hence, the estimator is efficient.

(4) Under the assumption that the noise power σ_w^2 is also unknown:

$\hat{A}_{ML} = \frac{1}{N} \sum_{n=0}^{N-1} (-1)^n x[n]$ is the same as before since it does not depend on the noise power, and so it has the same bias and MSE as before.

To derive the ML estimate of σ_w^2 we have to derive the log-likelihood function w.r.t. σ_w^2 :

$$\ln L(A, \sigma_w^2) = -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln(\sigma_w^2) - \frac{1}{2\sigma_w^2} \sum_{n=0}^{N-1} (x[n] - (-1)^n A)^2 = -\frac{N}{2} \ln(2\pi) - \frac{N}{2} \ln(\sigma_w^2) - \frac{1}{2\sigma_w^2} \|\mathbf{x} - A\mathbf{s}\|^2$$

$$\frac{d \ln L(A, \sigma_w^2)}{d\sigma_w^2} = -\frac{N}{2\sigma_w^2} + \frac{1}{2\sigma_w^4} \sum_{n=0}^{N-1} (x[n] - (-1)^n A)^2 = 0 \Rightarrow \hat{\sigma}_{wML}^2 = \frac{1}{N} \sum_{n=0}^{N-1} (x[n] - (-1)^n A)^2.$$

Since $\hat{A}_{ML} = \frac{1}{N} \sum_{n=0}^{N-1} (-1)^n x[n]$, we finally get:

$$\hat{\sigma}_{wML}^2 = \frac{1}{N} \sum_{n=0}^{N-1} (x[n] - (-1)^n \hat{A}_{ML})^2 = \frac{1}{N} \sum_{n=0}^{N-1} \left(x[n] - (-1)^n \frac{1}{N} \sum_{k=0}^{N-1} (-1)^k x[k] \right)^2$$

$$E \{ \hat{\sigma}_{wML}^2 \} = E \left\{ \frac{1}{N} \sum_{n=0}^{N-1} (x[n] - (-1)^n \hat{A}_{ML})^2 \right\} = \frac{1}{N} \sum_{n=0}^{N-1} E \left\{ (x[n] - (-1)^n \hat{A}_{ML})^2 \right\}$$

$$\begin{aligned} E \left\{ (x[n] - (-1)^n \hat{A}_{ML})^2 \right\} &= E \{ x^2[n] \} + E \left\{ (-1)^{2n} (\hat{A}_{ML})^2 \right\} - 2(-1)^n E \{ x[n] \hat{A}_{ML} \} \\ &= E \{ x^2[n] \} + E \left\{ (\hat{A}_{ML})^2 \right\} - 2(-1)^n E \{ x[n] \hat{A}_{ML} \} \end{aligned}$$



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$$E\{x^2[n]\} = (E\{x[n]\})^2 + \text{var}\{x[n]\} = (-1)^{2n} A^2 + \sigma_w^2 = A^2 + \sigma_w^2$$

$$E\{(\hat{A}_{ML})^2\} = (E\{\hat{A}_{ML}\})^2 + \text{var}\{\hat{A}_{ML}\} = A^2 + \frac{\sigma_w^2}{N}$$

$$\begin{aligned} E\{x[n]\hat{A}_{ML}\} &= E\left\{x[n]\left(\frac{1}{N}\sum_{k=0}^{N-1}(-1)^k x[k]\right)\right\} = \frac{1}{N}\sum_{k=0}^{N-1}(-1)^k E\{x[n]x[k]\} \\ &= \frac{1}{N}\sum_{k=0}^{N-1}(-1)^k \left((-1)^k (-1)^n A^2 + \sigma_w^2 \delta[k-n]\right) = \frac{1}{N}(-1)^n A^2 \sum_{k=0}^{N-1}(-1)^{2k} + \frac{1}{N}\sum_{k=0}^{N-1}(-1)^k \sigma_w^2 \delta[k-n] \\ &= (-1)^n A^2 + (-1)^n \frac{\sigma_w^2}{N} = (-1)^n \left(A^2 + \frac{\sigma_w^2}{N}\right) \end{aligned}$$

$$\begin{aligned} E\{\hat{\sigma}_{ML}^2\} &= \frac{1}{N}\sum_{n=0}^{N-1}\left(A^2 + \sigma_w^2 + A^2 + \frac{\sigma_w^2}{N} - 2(-1)^n (-1)^n \left(A^2 + \frac{\sigma_w^2}{N}\right)\right) \\ &= \frac{1}{N}\sum_{n=0}^{N-1}\left(2A^2 + \sigma_w^2 + \frac{\sigma_w^2}{N} - 2A^2 - 2\frac{\sigma_w^2}{N}\right) = \sigma_w^2 - \frac{\sigma_w^2}{N} \\ &= \sigma_w^2 \left(1 - \frac{1}{N}\right) \neq \sigma_w^2 \end{aligned}$$

The bias is given by:

$$E\{\sigma_w^2 - \hat{\sigma}_{ML}^2\} = \sigma_w^2 - \sigma_w^2 \left(1 - \frac{1}{N}\right) = \frac{\sigma_w^2}{N}.$$

Hence, the estimator is biased but asymptotically unbiased.